Yashodhan Deepak Hakke

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Education

Virginia Tech — **Aug 2024 – May 2026**
Master of Science in Computer Engineering — *Blacksburg, VA, USA*

- **Relevant Coursework**: Computer Vision, Advanced Machine Learning, Stochastic Signals, Quantum Engineering, Computer Architecture, Network Architecture & Protocols.

Savitribai Phule Pune University — **Aug 2020 – May 2024**
Bachelor of Technology in Electronics Engineering — *Pune, India*

- **Relevant Coursework:** Digital Signal Processing, Embedded Systems, Microcontrollers, VLSI Design, Control Systems, Computer Networks, Signals & Systems, Power Electronics, Operating Systems, Deep Learning.

Work Experience

Cinemind | — **May 2025 – Present**
Software Engineer

- Shipped an AI Script Coverage platform utilized by film/TV teams **(incl. Universal Pictures, Amblin Entertainment etc)**, automating screenplay analysis end-to-end.
- Engineered NLP pipeline using RAG to process 600+ page screenplays into automated scene breakdowns and sentiment reports.
- Architected a constraint-satisfaction **Scheduling Engine (beta)** that cut planning time by **99% (3 weeks to under 4 hours).**
- **Stack:** Python (LLM orchestration, OR-Tools), TypeScript, Next.js, LangChain, Docker, AWS (EC2, S3, IAM), OpenAI API, Google Cloud Platform, PostgreSQL, Stripe, GitHub.

Bradley Dept. of Electrical and Computer Engineering, Virginia Tech — **Oct 2024 – May 2025**
Graduate Research Assistant (Fully Funded) — *Blacksburg, VA*

- Built a 12-state Cyber-Physical-Social simulator modeling fear propagation, fake news dynamics, and EMS allocation for community resilience.
- Developed a **control-affine** actor–critic controller ($\dot{x} = f(x) + G(x)u + K(x)d$) to minimize social volatility under bounded interventions.
- Optimized batched rollouts to scale Multi-Agent RL training across a population of 10,000+ simulated entities, reducing peak panic by 30%.
- Validated trends on Hurricane Harvey signals (Twitter/FEMA); invited speaker at **Columbia University (NESCW'25)**.
- **Stack:** Python, PyTorch, Github, CUDA, Linux.

Devot AI — **Nov 2023 – Jul 2024**
Machine Learning Engineer Intern — *Bengaluru, India*

- Improved LLM reasoning by **12%** through RLHF-style evaluation and fine-tuning on **300+** user prompts.
- Engineered **1,000+** web-search test cases to optimize tool-use logic, reducing retrieval hallucinations by **25%**.
- Architected Python ETL pipelines for Amazon Ads, cutting reporting latency by **88%** (4 hrs to 30 mins) for 10+ clients.
- **Stack:** Python, Pandas, SQL, OpenAI API, RLHF, ETL, Automated Testing.

Ankureto Store — **Oct 2021 – Jan 2022**
Full Stack Engineer Intern — *Pune, India*

- Built a custom React e-commerce platform with REST APIs, processing 150+ orders in the first 2 months post-launch.
- Optimized frontend asset delivery and rendering to reduce page load times by 40%, driving a 20% increase in sales conversion during peak traffic periods.
- Deployed a responsive web application on AWS via automated CI/CD pipelines, ensuring high availability with 99%+ service uptime.
- **Stack:** React.js, JavaScript, HTML/CSS, AWS, REST APIs.

Projects

Agentic Browser (Veena.ai) | Live Demo — **Apr 2025 – May 2025**

- Built a Perplexity-style search engine utilizing Groq LPUs to stream cited answers 3x faster than standard GPT-4.
- Designed a low-latency streaming UX on Next.js/Vercel and validated adoption by onboarding **10+ live users**.
- **Stack:** Next.js, Vercel, RAG, SerpAPI, Tavily, OpenAI, Groq, Embeddings.

GraphDocuMind | Github — **Feb 2025 – Mar 2025**

- Built a GraphRAG pipeline converting PDFs into Neo4j knowledge graphs to boost retrieval precision vs. vector search.
- Integrated LlamaParse + LlamaIndex to generate verifiable citations grounded in graph-linked document entities.
- **Stack:** Python, Neo4j, LlamaIndex, GraphQL, Llama3, Docker.

ChalBeyy (Ride Sharing) | Live Website — **Jan 2025 – Mar 2025**

- Architected PostGIS-powered carpool matching with <2s latency, supporting 25+ concurrent active users.
- Designed a Node.js+Redis microservices backend on AWS with REST APIs powering a responsive, high-performance React UI.
- **Stack:** Node.js, Express, PostgreSQL (PostGIS), Redis, React, Docker.

Lagori Robot (ABU Robocon) | Video Link — **Aug 2021 – Jun 2022**

- Built the actuation/control stack for two ABU Robocon 2022 Lagori robots, enabling reliable hit execution in match play.
- Led actuation and system integration across Raspberry Pi + Arduino + ESP32, and prototyped vision-based targeting with OpenMV+YOLOv5 with 95% accuracy.
- **Stack:** Python, C++, Raspberry Pi, Arduino, ESP32, OpenMV, YOLOv5.

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS, Bash, MATLAB
Frameworks: React.js, Vue.js, Next.js, Node.js, Express, Django, FastAPI, PyTorch, TensorFlow, LangChain
Cloud & DevOps: AWS (EC2, Lambda, S3), GCP, Docker, Kubernetes (containerization & orchestration), Terraform (Infrastructure as Code), CI/CD (GitHub Actions), Linux, Git
Databases: SQL (PostgreSQL), NoSQL (MongoDB, Redis), Neo4j, Elasticsearch
Tools: Postman, VS Code, Cursor, Claude Code, Vercel, NVIDIA TensorRT
Concepts: Object-Oriented Programming (OOP), Data Structures and Algorithms, Graph Theory, Game Theory, Backend Development, Full-Stack Development, Distributed Systems, System Design, Microservices, RESTful APIs, GraphQL APIs, RAG, LLMs, Agile